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Mind, Brain, and Meaning

Short Paper 1

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Landing on Marr's

Marr's levels of analysis, widely taught in introductory cognitive science and psychology courses, are commonly criticized because of its separation of seemingly inextricably linked processes and modes of analysis. I've personally seen students have lengthy conversations with professors, confused about the difference between Marr's first and second levels, computational and algorithmic, and frustrated when the answer comes to show the enormous overlap between them. While the distinction between some of Marr's levels may seem fuzzy at first, each level does have a specific purpose when analyzing information processing systems: the computational level identifies the goal/aim of a task, the algorithmic level looks at the algorithm, or steps, by which inputs can be transformed to reach the goal, and the implementational level analyzes how that algorithm and goal can be physically realized (Marr, 25). Considering these definitions of the levels, it's hard to imagine any of the levels standing alone - each level requires the other two to explain how any information task is completed. Viewing mental phenomena like perception, thought, feeling, or memory as information processing tasks, and considering the criteria for a "satisfying" explanation of these tasks - having a clear understanding of the task (A), seeking a deep understanding of the physical nature of the brain (B), and being able to connect these two understandings (C) - the overlap in Marr's proposed three levels actually makes his approach to the brain as an information processing system a rather apt and promising one in understanding mental phenomena.

Marr's interconnected levels help us avoid reductionist arguments that solely attribute our understanding of mental phenomena, like feeling, to neurological findings. The relationship between the algorithmic and computational levels especially help us to fulfill criteria C by showing the deep interconnectedness of the physical nature and overall goal of mental processes. In Miller and Keller's discussion on the faulty reductions of findings in psychology to neuroscientific findings, they discuss why an emotion like fear is irreducible to biology or neurological study: “Because “fear” means more than a given type of neural activity, the concept of fear is not reducible to neural activity. Researchers are learning a great deal about the biology of fear—and the psychology of fear—from studies of the amygdala…but this does not mean that fear is activity in the amygdala. That is simply not the meaning of the term” (Miller & Keller, 212). Viewing the fear emotion as an information processing task - there was presumably some input that was transformed into an emotional response, fear, as its output - a computational analysis of fear according to Marr would presumably try and analyze a question about why fear was the output given some input or what about the inputs made fear the goal response. This understanding would satisfy criteria A because it would give us an initial understanding of what fear is and why we respond with it. An implementational level of fear would look at the synapses in our brain when fear is physically or self-reportedly exhibited in a person and would likely direct our attention to the amygdala as being the controller of our fear response. This would effectively fulfill criteria B. Criteria C is fulfilled when we realize that fear may only be a response we can have because of the faculties of our brain, like the amygdala, or that connections in the amygdala are only strengthened because we tend to have this response/goal often as humans (Lecture 1, Slide 36). It is hard for me to evaluate which of these is more true, but I believe both lines of reasoning are plausible and help us understand that either form of understanding can be informed by the other. Miller and Keller echo this point succinctly at the end of their piece: If one views brain tissue as implementing psychological functions, the expertise of cognitive science is needed to characterize those functions, and the expertise of neuroscience is needed to study their implementation. Each of those disciplines will benefit greatly from the other, but neither encompasses, reduces, or underlies the other (Miller & Keller, 215).

Marr’s approach to treating the mind as an information processing system seems further promising when being able to apply the three levels of analysis to 17th-century thinker Descartes exploration of the connection between the mind and body and dualistic existence of himself in his chapter titled “Sixth Meditation”. Criteria A and Criteria B of our necessary definition of satisfying mental phenomena somewhat correspond, respectively, to mind and body. As previously explored, the computational level of analysis helps to satisfy Criteria A. In Descartes' text, he explores his ability to engage in thought and belief, essentially trying to understand what is true given that he is thinking these thoughts. His exploration of the nature of his mind is a stab at understanding what is true and what thinking really is. Satisfying criteria B, analyzing Descartes through the implementational level, Descartes posits that there is a physical nature and creator to his ideas and thoughts: “on the contrary, he has given me a great propensity to believe that they are produced by corporeal things” (Descartes, 63). Criteria C is actually quite beautifully satisfied by the remaining level of analysis, the algorithmic level, in Descartes’s writing. As Descartes discusses the existence of a corporeal or extended substance that informs his intellectual acts, he says, “This substance is either a body, that is, a corporeal nature, in which case it will contain formally <and in fact> everything which is to be found objectively <or representatively> in the ideas; (Descartes, 62). Even Descartes' language, “representatively”, corresponds to Marr’s second level of analysis: the algorithmic or representational level. In class, we discussed that Descartes believed the pineal gland housed the soul; the idea that some physicality and the mind interact to form his own intellect or the soul is an algorithmic description because it insinuates that the body helps to transform information that facilitates a mental process. To lay it out plainly, the algorithmic description would be that the system, Descartes, is able to transform information through a corporeal faculty (Lecture 1, Slide 29).

While Descartes’s writings come to us much before Marr’s, Marr’s levels of analysis give us a structure to better understand Descartes mind-body dualism even though Descartes likely didn’t think of the mind as an information processing system. If Marr’s levels of analysis can both satisfy the criteria to adequately explain mental phenomena and bridge our understanding of the mind today with a 17th-century understanding of the mind, then treating the brain as an information-processing system seems rather promising.

Works Cited

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